

DATA EXPLORATION

AMIRTHARAJ R

DATA ANALYTICS EXCEL ASSIGNMENT - 2

Handling Missing Values						
?	Check for missing values in the 'Price' column. How would you handle products with missing price information?					
Ans	Answer worked out in E Column					
?	If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.					
Ans	Categories are imputed based on the products and features Ex: Adidas = It was a Fashion Wear					
Correcting Inconsistent Data:						
?	Identify any inconsistent text formats present in the "Product Name" column.					
Ans	laptop			Laptop		
?	Identify any typos present in the "Category" column.					
Ans	Electroni			Electronics		
?	Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.					
Ans	Products Name Modified using "Proper" Formula. In Category Electroni replaced to Electronics					Electronics
Removing Duplicates:						
?	Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.					
Ans				Duplications removed by using "Product ID" Column		
Splitting and Merging Data:						
?	Split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.					
Ans	28-JAN-US	Manufacturing Date	28	28-JAN-US	Country Code	US
?	Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".					
Ans	Product Name	Laptop	Brand Name	Dell	Product Brand	Laptop Dell
Number Formatting						
?	Format the data type of the "Price" column to currency format					
Ans	Price	1000	Price	₹ 1,000.00		
?	Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY" format					
Ans	Manufacturing Date	28-JAN-US	DD-MM-YYYY	28-01-2026		
Conditional Formatting						
?	Apply data bar or color scales conditional formatting in the "Price" column.					
Ans	Price	1000	Price	₹ 1,000.00		
?	Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."					
Ans	Category	Electronics	Category	Electronics		

Find and Replace

Find

Replace

Find what:

Electroni

Replace with:

Electronics

Options >>

Replace All

Replace

Find All

Find Next

Close

Product ID	Product Name	Brand Name	Price (\$)	Q 1	Quantity	Category
28-JAN-US	Laptop	Dell	1000	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	900	25	Electronics
22-MAY-US	Backpack	North Face	70	70	20	Fashion
07-JUN-UK	Headphones	Sony		309.3548387	45	Electronics
19-JUL-UK	T-Shirt	Adidas	30	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	120	35	Electronics
18-FEB-CA	Smartwatch	Fitbit	150	150	15	Electronics
16-APR-ES	Headphones	Bose	250	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	150	15	Fashion
14-AUG-US	Camping Tent	Coleman		309.3548387	10	Outdoor
14-MAY-RU	Camera	Nikon	700	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban		309.3548387	25	Fashion
11-JUL-CA	Blender	Vitamix	400	400	40	Kitchen
16-APR-ES	Headphones	Bose	250	250	20	Electronics
07-MAR-CA	Dress	Zara	60	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	50	Fashion
17-JUN-IN	Laptop	HP	950	950	25	Electronics
09-JUL-FR	Watch	Casio	100	100	20	Accessories

Conclusion:

In conclusion, this Excel assignment successfully demonstrated the key steps involved in **data cleaning and preparation** of the Product Dataset. Missing values were identified and appropriately handled, while inconsistent product names and category typos were standardized using Excel functions and Find & Replace. Duplicate records were removed, and the **Product ID** was transformed into separate **Manufacturing Date** and **Country Code** columns. The **Brand Name** and **Product Name** were also merged into a new **Product Brand** column. In addition, suitable currency and date formats were applied, and conditional formatting was used to highlight price patterns and the **Electronics** category. These cleaning and transformation steps improved the **accuracy, consistency, readability, and usability of the dataset**, making it more suitable for further data analysis and visualization.

This conclusion matches the completed work shown in your Word document, including the Find & Replace, duplicate removal, column splitting/merging, formatting, and conditional-formatting results.